

Is It Still a U-shape? Empirical Study on the Dynamics of Reward-Based Crowdfunding under Mitigated Uncertainty

Abstract

Potential backers at reward-based crowdfunding must make funding decisions when faced with uncertainty due to information asymmetry about crowdfunding project qualities. Potential backers imitate other backers who have already provided funds to overcome uncertainty. The U-shape dynamic patterns of crowdfunding backers' behaviors under this uncertainty has allowed entrepreneurs to establish effective strategies. However, potential backers do not need to wait for others' decisions in mitigated uncertainty settings where a reliable quality signal exists. Therefore, our study aims to analyze how the award effect changes the initial and ending dynamics. We believe that a crowdfunding project run by award-winning entrepreneurs would create strong L-shape dynamics. Further, we utilize the difference-in-differences approach as our main empirical analysis to examine the changes in crowdfunding projects' initial and ending dynamics after the awarding. This research will help entrepreneurs to understand backers' behavior over the crowdfunding cycle under less uncertain situations. The results indicate that entrepreneurs should take a different strategy from normal crowdfunding after winning an award.

Keywords: crowdfunding, U-shape dynamics, award effect, quality signal

1. Introduction

A challenge for business ventures is securing early-stage funding (Cosh et al. 2009). So, many entrepreneurs have experienced difficulties soliciting initial financial support, as most venture capitalists prefer to invest in firms at the growth stage (Grant 2013, Roma et al. 2018). Alternative to traditional fundraising methods (e.g., venture capital funds and angel investors), entrepreneurs can now rely on crowdfunding, which is a widely used vehicle for raising funds by tapping many backers on an online platform (e.g., Kickstarter, Indiegogo, Wadiz, etc.) (Bruton et al. 2015, Mollick 2014). The size of the global crowdfunding market was estimated at \$13.64 billion in 2021; further, it was predicted that the market size will approach \$28.92 billion in 2028 (Statista 2022).

There are four types of crowdfunding: reward-based crowdfunding, lending-based crowdfunding, donation-based crowdfunding, and equity-based crowdfunding (Cumming 2014). Among of them, reward-based crowdfunding follows an all-or-nothing fundraising mechanism. Entrepreneurs receive funding, while backers receive the reward (e.g., goods or services) only when the funding goal is achieved at the target goal amount (Li et al. 2020). Potential backers must make a funding decision under uncertainty due to information asymmetry about campaign qualities (Chan et al. 2020, Colombo et al. 2015). Uncertainty about crowdfunding success and reward quality can lead

potential backers to observe and imitate other backers' behaviors (e.g., rational herding behavior) who have already funded (Vismara 2016, Zhang and Liu 2012). However, when there is a lower level of uncertainty, it is questionable whether potential backers will still follow prior backers' behaviors.

The literature has empirically demonstrated the U-shape dynamics of crowdfunding backers' behaviors over the crowdfunding cycle (Chan et al. 2020, Kuppuswamy and Bayus 2018). The funding amount decreases at the initial stage and increases at the ending stage (Chan et al. 2020, Kuppuswamy and Bayus 2018). Additionally, fundraising is stuck in the middle stage (Ordanini et al. 2020). Thus, the U-shape is formed between the crowdfunding cycle and the funding amount (Chan et al. 2020, Kuppuswamy and Bayus 2018). The U-shape dynamics allow entrepreneurs to establish effective crowdfunding operation strategies (Li et al. 2020). According to Li et al. (2020), the success of a crowdfunding project is closely related to the promotion strategies, such as the timing of the promotion and the intensity of efforts. Thus, the literature on U-shape dynamics has mainly focused on how potential backers make a decision when facing uncertainty regarding reward quality. However, we note that an entrepreneur's winning award could mitigate a backer's uncertainty. As such, we believe that backers would show a different dynamic pattern under low uncertainty compared with the U-shape dynamic one under high uncertainty.

Thus, this study will investigate the different dynamic patterns of crowdfunding backers' behaviors over the crowdfunding cycle under mitigated uncertainty settings. Under the reduced uncertainty, we study whether there is still a U-shape relationship between prior funding and subsequent contributions and how backer numbers and fundraising change over time. Moreover, we will estimate how initial and ending effects change over the crowdfunding cycle due to reliable quality signals. Our study examines this research question in the context of Wadiz, South Korea's leading reward-based crowdfunding platform. We will compare the dynamic pattern of a crowdfunding project run by award-winning entrepreneurs and other crowdfunding projects (i.e., the most common crowdfunding without the award). Among those awarded entrepreneurs' projects, we analyze the award-winning effects by splitting them into two types: the same product with award-winning crowdfunding (e.g., receive an crowdfunding award for a cleanser and rerun crowdfunding with the same cleanser) and the different product with award-winning crowdfunding projects (e.g., receive a crowdfunding award for a cleanser and run crowdfunding with another product such as lotion). In addition, we examine the effect of the encore (i.e., rerunning past successful crowdfunding with the same reward), which acts as a quality signal similar to the awards. According to the previous studies in award effects (Foerderer et al. 2021, Frey and Gallus 2017), we assume that the award-winning effect could reduce the uncertainty of crowdfunding projects. Thus, potential backers don't need to wait for another's decision and thus make a decision in the early stage of crowdfunding projects. As such, we believe that the dynamics of a crowdfunding project run by award-winning entrepreneurs would result in strong L-shape dynamics.

We adopt the difference-in-differences (DiD) approach as our main empirical analysis to examine how crowdfunding projects' initial and ending dynamics change after the awarding. We also apply propensity score matching (PSM) to split a crowdfunding project into treatment control groups, depending on whether or not the entrepreneur is awarded. The crowdfunding project of the award-winning entrepreneur is the treatment group, while the crowdfunding project of the nonawarded entrepreneur is the control group. We conduct parallel trend assumption and falsification tests to assure no trend difference between treatment group and control group before announcing the award.

The research takes a step further from prior literature by analyzing the dynamics of crowdfunding backers' behaviors over the crowdfunding cycle under less uncertain situations. Dynamics of award-winning entrepreneur's crowdfunding might indicate different patterns from the dynamics of initial crowdfunding that prior researchers have already studied. The results identify that, given the reduced uncertainty of the backers, entrepreneurs should take a different strategy from that of normal crowdfunding.

2. Literature Review

2.1. Crowdfunding

2.1.1. Project-Level Research of Crowdfunding

Reward-based crowdfunding is the preferred fundraising method for individuals and entrepreneurs to start their businesses. Early literature has focused on finding critical factors of crowdfunding success. Social connection is a key factor in achieving target fundraising. Notably, close social relations' (e.g., family and friends) fundraising is important (Agrawal et al. 2015, Steinberg 2012). Entrepreneurs attract a large portion of the fundraising from those indirectly connected through social network services, along with people they know directly (Kuppuswamy and Bayus 2018). Thus, the relationship between the entrepreneurs' social media activities for crowdfunding and the funding amount is closely related (Thies et al. 2016).

Quality signals also have a positive effect on crowdfunding success. In reward-based crowdfunding platforms, potential backers make decisions without enough information to gauge the quality of the crowdfunding project (Chan et al. 2020, Colombo et al. 2015, Vismara 2016, Zhang and Liu 2012). For example, potential backers do not know whether an entrepreneur is in the conceptual stage, development stage, or the stage where a product has already been created (Mollick 2014, Vismara 2016). Thus, few quality signals act as valuable information for potential backers. Preparedness is among the quality signals that increase the chance of crowdfunding success (Mollick 2014). Moreover, an entrepreneur can use posting updates at the platform about funding events or new developments as a quality signal (Hornuf and Schwienbacher 2018). According to Kunz et al. (2017), to increase the probability of reward-based crowdfunding success, an entrepreneur should make a significant effort to communicate with potential backers.

2.1.2. Dynamic Research of Crowdfunding

Prior literature about dynamic patterns of crowdfunding backers' behaviors over the crowdfunding cycle has added an empirical understanding of reward-based crowdfunding. In an asymmetric information situation, potential backers tend to follow others' decisions (i.e., rational herding behavior) who have already raised funds (Chan et al. 2020, Colombo et al. 2015, Vismara 2016, Zhang and Liu 2012). Thus, early research has focused on the relationship between prior funding and subsequent contributions. The crowdfunding pattern over the funding cycle, which can be seen as the funding period and the prior fundraising amount, is U-shape (Chan et al. 2020, Kuppuswamy and Bayus 2018). Potential backers join more in the initial or ending period of a project than in the middle (Kuppuswamy and Bayus 2018). As Chan et al. (2020) stated, subsequent fundraising, according to the level of prior funding, is also U-shape. Subsequent contributions, which are daily paid amounts, decrease at a low level of prior funding and increase at a high level of prior funding (Chan et al. 2020).

Prior researchers have revealed additional findings. The crowdfunding pattern over the funding cycle is U-shape regardless of success, target fundraising, and project category (Kuppuswamy and Bayus 2018). Furthermore, Kuppuswamy and Bayus (2018) identified that close social relations support initial and ending periods. Additionally, backing propensity and promotion strategies rely on positive externalities and negative deadline effects (Li et al. 2020). Previous studies have analyzed the crowdfunding dynamics by aggregating. However, each crowdfunding campaign may have a different degree of uncertainty that potential backers recognize. For instance, potential backers may feel more confident about crowdfunding projects that have award-winning experience than crowdfunding that does not. Hence, to our best knowledge, there is little study on the dynamics of crowdfunding projects with different levels of uncertainty.

2.2. Awards

Awards are the symbol to praise outstanding achievements worthy of recognition (Frey and Gallus 2017). The most important advantage of awards is gaining social status that results in a visible distinction from others (Frey and Gallus 2017). The two features of the award, information content and observability, allow the award to serve as a quality signal which can reduce uncertainty in information asymmetries (Foerderer et al. 2021). The platform owner selects only a few crowdfunding; awards could be reliable quality signals (Foerderer et al. 2021).

Previous studies on the dynamics of crowdfunding failed to distinguish between normal crowdfunding and award-winning entrepreneurs' crowdfunding projects, despite differences in uncertainty perceived by potential backers. In addition, researchers have yet to analyze the dynamic patterns of crowdfunding backers' behaviors in settings where awards have reduced crowdfunding quality risks. Specifically, prior literature understudied the changes of initial or ending effect under reduced uncertainty situations. Our research thus aims to analyze the dynamic patterns of crowdfunding due to award-winning effect and indicate the effect of reliable quality signals in the context of crowdfunding.

3. Research Design

3.1. Data

The research context of our study is Wadiz, a leading reward-based crowdfunding platform in South Korea. The entrepreneur ran 11,512 crowdfunding, and 551,581 backers funded more than 162 billion KRW in 2021. We selected Wadiz as our research context, as the platform confers awards to outstanding entrepreneurs and crowdfunding annually. The platform selects award-winning entrepreneurs and crowdfunding by reviewing backers' satisfaction, achievement amount, number of backers, active communication, etc. Also, the platform publicly reveals award and funding status. So, anyone can see the award-winning crowdfunding list, award type, and year of the award. In addition, the platform provides crowdfunding information in real-time, such as fundraising, achievement rate, number of sponsors, number of likes, and number of Facebook shares.

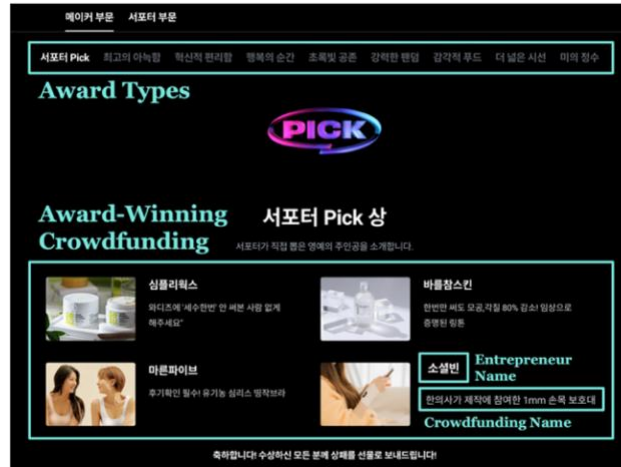


Figure 1. Award Types and Award-Winning Crowdfunding List

The platform announces award-winning crowdfunding projects for several award types each year (Figure 1). For instance, platform users can observe nine types of awards on the platform's 2021 awards page (e.g., backers' pick, greatest coziness, innovative convenience, powerful fandom, etc.). For each award type, four to twenty-three entrepreneurs receive awards. The awards page also offers thumbnails, entrepreneur names, and crowdfunding names. Platform users can jump to an entrepreneur's page by clicking either the entrepreneur or crowdfunding name.



Figure 2. Example of Entrepreneur Page and Crowdfunding Project Details

The entrepreneur page delivers a list of an entrepreneur's ongoing and closed crowdfunding. Platform users can jump to the crowdfunding project page to see fundraising status. As shown in Figure 2, the crowdfunding project details page offers real-time information on crowdfunding status (e.g., fundraising, achievement rate, number of sponsors, etc.) and evaluation data on entrepreneurs (e.g., satisfaction score, average response time, number of total past backers, etc.). We will collect data from November 1, 2020, to November 1, 2022.

3.2. Variables

Variables	Description
Daily Backers	Daily number of backers added
Daily Amount	Daily amount of fundraising gathered

Daily Percentage	Daily fundraising achievement rate (total fundraising amount over target fundraising amount)
Entrepreneur Award	Dummy variable indicates whether it is award-winning entrepreneur's crowdfunding project (0/1)
After Award	Dummy variable indicates whether the observation is after the award announcement date (0/1)
Crowdfunding Award	Dummy variable indicates whether it is award-winning crowdfunding project rewards (0/1)
Crowdfunding Encore	Dummy variable indicates whether it is re-running crowdfunding with the same product (0/1)
Crowdfunding Elapsed Time	Date elapsed from the funding start date
Cycle ^{period}	Dummy variable indicates the period of a crowdfunding cycle (initial/middle/ending)
Forged After Award	Dummy variable 0 for up to five days before the award announcement date and 1 for five days after
Crowdfunding Period	Duration of funding
Crowdfunding Goal	Target fundraising amount
Crowdfunding Likes	Total number of likes gathered
Crowdfunding Referrals	Total number of times users have shared the crowdfunding on Facebook
Crowdfunding Reward Price	Average price of crowdfunding rewards
Entrepreneur Satisfaction Level	Entrepreneur's satisfaction score for past funding
Entrepreneur Followers	Number of users who follow the entrepreneur on platform

Table 1. Variables and Description

Our study aims to analyze changes in initial and ending dynamics due to the award effect. Table 1 indicates the variable description for our future analysis. We use the number of backers as the main dependent variable. Using the platform's yearly award list, we construct a dummy variable for difference-in-difference (DiD) analysis. The value of the dummy variable is 1 for the crowdfunding project conducted by the award-winning entrepreneur and 0 otherwise. Additionally, we construct a categorical variable to analyze award effects by dividing them into initial, middle, and ending periods. We calculate the percentage of the crowding project period with a value between 0 and 1 for each crowdfunding project. Percentage of the crowdfunding project is measured by the elapsed days over the total fundraising period. We create the categorical variable by classifying the 0% to 25% section as the initial period, 25% to 75% section as the middle period, and 75% to 100% section as the ending period. Moreover, the dummy variable $AfterAward_{it}$ is 1 if the observation is after the award announcement date and 0 otherwise. The dummy variable $CrowdfundingAward_i$ is 1 if a crowdfunding project raises the same reward as an award-winning project by an award-winning entrepreneur, or 0 if not.

3.3. Methodology

We examine the changes in crowdfunding projects' initial and ending dynamics after the awarding. We use the difference-in-differences approach as our main empirical analysis to measure the effect of treatment while eliminating the heterogeneity between the treatment and control groups. We set winning the 2021 Wadiz Awards as a treatment of DiD analysis. The treatment allows us to analyze the award effect by measuring changes in the number of crowdfunding project backers before and after the winning award. We define the treatment group as a crowdfunding project by an award-winning entrepreneur and the control group as a crowdfunding project conducted by a nonawarded entrepreneur.

To mitigate the selection bias and guarantee the homogeneity of the treatment group and control group attributes, we apply our DiD analysis combined with the propensity score matching (PSM) methodology (Imbens and Woodridge 2009, Kumar et al. 2018). Further, we use PSM to identify control crowdfunding projects similar to award-winning entrepreneurs' crowdfunding projects. We will check whether the treatment and control groups follow similar trends through the parallel trend assumption test. Furthermore, our study conducts the falsification test to ensure no trend difference between the awarded and nonawarded projects before the actual date of award granting. Model specification for the falsification analysis is as follows:

$$\begin{aligned} \text{DailyBackers}_{it} = & \alpha + \beta_1 \text{EntrepreneurAward}_i + \beta_2 \text{ForgedAfterAward}_{it} \\ & + \beta_3 (\text{EntrepreneurAward}_i \times \text{ForgedAfterAward}_{it}) + \varphi' \text{Controls}_{it} + e_{it} \end{aligned}$$

Specifically, we use a forged treatment date by bringing forward the actual treatment date by five days of all projects and checking if the estimates of our DiD terms are statistically insignificant. Models compare the trends from one to five days to the baseline trends from six to ten days, both observed before the actual treatment date. The dependent variable DailyBackers_{it} is the daily number of backers added to crowdfunding project i on day t . $\text{EntrepreneurAward}_i$ is a dummy variable set to 1 if a crowdfunding project i is run by an award-winning entrepreneur and 0 otherwise. $\text{ForgedAfterAward}_{it}$ dummy variable assumes that the pretended treatment date is five days before the actual treatment date. The variable's value is 1 for the date before the pretended treatment date and 0 for the date after. β_3 is the main parameter indicating trend differences.

4. Future Plan

We will collect additional observations to implement PSM more accurately. Moreover, we will study the effect of awards on the number of crowdfunding project backers in a DiD setting. We also plan to conduct various DiD and PSM analyses to obtain reliable results. In addition, our research will clarify whether the award-winning effect is limited to award-winning crowdfunding project rewards or even different products of award-winning entrepreneurs. We will split the crowdfunding project that the entrepreneur runs with the same rewards as the award-winning project and the project with the different rewards from the award-winning project after winning the award. Moreover, we will offer more theoretical arguments for our study in further research.

References

References are available upon request.